

```
In [ ]: #Kaggle task for datascience with Pandas ai
# https://www.kaggle.com/code/harunshimanto/pandas-with-data-science-ai
```

```
In [1]: import pandas as pd          #import libraries
```

```
In [7]: ratings = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\archive\rating.csv')
```

```
In [9]: ratings.shape
```

```
Out[9]: (20000263, 4)
```

```
In [10]: ratings
```

```
Out[10]:
```

	userId	movieId	rating	timestamp
0	1	2	3.5	2005-04-02 23:53:47
1	1	29	3.5	2005-04-02 23:31:16
2	1	32	3.5	2005-04-02 23:33:39
3	1	47	3.5	2005-04-02 23:32:07
4	1	50	3.5	2005-04-02 23:29:40
...	...	...	...	...
20000258	138493	68954	4.5	2009-11-13 15:42:00
20000259	138493	69526	4.5	2009-12-03 18:31:48
20000260	138493	69644	3.0	2009-12-07 18:10:57
20000261	138493	70286	5.0	2009-11-13 15:42:24
20000262	138493	71619	2.5	2009-10-17 20:25:36

20000263 rows × 4 columns

```
In [11]: ratings.head(3)
```

```
Out[11]:
```

	userId	movieId	rating	timestamp
0	1	2	3.5	2005-04-02 23:53:47
1	1	29	3.5	2005-04-02 23:31:16
2	1	32	3.5	2005-04-02 23:33:39

```
In [12]: tags = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\archive\tag.csv')
```

```
In [13]: tags
```

```
Out[13]:
```

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18
...	...	...	...	...
465559	138446	55999	dragged	2013-01-23 23:29:32
465560	138446	55999	Jason Bateman	2013-01-23 23:29:38
465561	138446	55999	quirky	2013-01-23 23:29:38
465562	138446	55999	sad	2013-01-23 23:29:32
465563	138472	923	rise to power	2007-11-02 21:12:47

465564 rows × 4 columns

```
In [14]: tags.shape
```

```
Out[14]: (465564, 4)
```

```
In [15]: tags.head(3)
```

```
Out[15]:
```

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19

```
In [16]: movies = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\archive\movie.csv')
```

```
In [17]: movies
```

```
Out[17]:
```

	movieId	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama Romance
4	5	Father of the Bride Part II (1995)	Comedy
...	...	...	...
27273	131254	Kein Bund für's Leben (2007)	Comedy
27274	131256	Feuer, Eis & Dosenbier (2002)	Comedy
27275	131258	The Pirates (2014)	Adventure
27276	131260	Rentun Ruusu (2001)	(no genres listed)
27277	131262	Innocence (2014)	Adventure Fantasy Horror

27278 rows × 3 columns

```
In [18]: movies.head(3)
```

```
Out[18]:
```

	movieId	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance

```
In [19]: movies.shape
```

```
Out[19]: (27278, 3)
```

```
In [20]: print(tags.columns)
print(ratings.columns)
print(movies.columns)
```

```
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
Index(['userId', 'movieId', 'rating', 'timestamp'], dtype='object')
Index(['movieId', 'title', 'genres'], dtype='object')
```

```
In [ ]: #Delete the timestamp columns as its not required.
```

```
In [21]: del tags['timestamp']
del ratings['timestamp']
```

```
In [22]: print(tags.columns)
print(ratings.columns)
print(movies.columns)
```

```
Index(['userId', 'movieId', 'tag'], dtype='object')
Index(['userId', 'movieId', 'rating'], dtype='object')
Index(['movieId', 'title', 'genres'], dtype='object')
```

```
In [23]: tags.head(3)
```

```
Out[23]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero

```
In [24]: tags.iloc[0]
```

```
Out[24]:
```

userId	18
movieId	4141
tag	Mark Waters

Name: 0, dtype: object

```
In [25]: tags.iloc[2]
```

```
Out[25]:
```

userId	65
movieId	353
tag	dark hero

Name: 2, dtype: object

```
In [27]: row_0 = tags.iloc[0]  
row_0
```

```
Out[27]:
```

userId	18
movieId	4141
tag	Mark Waters

Name: 0, dtype: object

```
In [28]: row_1 = tags.iloc[1]  
row_1
```

```
Out[28]:
```

userId	65
movieId	208
tag	dark hero

Name: 1, dtype: object

```
In [29]: row_0.index
```

```
Out[29]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [30]: row_0['userId']
```

```
Out[30]: np.int64(18)
```

```
In [31]: 'rating' in row_0
```

```
Out[31]: False
```

```
In [32]: row_0.name
```

```
Out[32]: 0
```

```
In [33]: row_0 = row_0.rename('firstRow')  
row_0.name
```

```
Out[33]: 'firstRow'
```

```
In [34]: row_0
```

```
Out[34]: userId      18  
movieId    4141  
tag        Mark Waters  
Name: firstRow, dtype: object
```

```
In [35]: row_1
```

```
Out[35]: userId      65  
movieId    208  
tag        dark hero  
Name: 1, dtype: object
```

```
In [ ]: #DataFrames
```

```
In [37]: tags.head()
```

```
Out[37]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

```
In [39]: tags.index
```

```
Out[39]: RangeIndex(start=0, stop=465564, step=1)
```

```
In [40]: tags.columns
```

```
Out[40]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [41]: tags.iloc[ [0,11,500] ]
```

```
Out[41]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
11	65	1783	noir thriller
500	342	55908	entirely dialogue

```
In [ ]: #Descriptive Statistics
```

```
In [42]: ratings['rating'].describe()
```

```
Out[42]: count    2.000026e+07
          mean     3.525529e+00
          std      1.051989e+00
          min      5.000000e-01
          25%      3.000000e+00
          50%      3.500000e+00
          75%      4.000000e+00
          max      5.000000e+00
          Name: rating, dtype: float64
```

```
In [43]: ratings.describe()
```

```
Out[43]:
```

	userId	movieId	rating
count	2.000026e+07	2.000026e+07	2.000026e+07
mean	6.904587e+04	9.041567e+03	3.525529e+00
std	4.003863e+04	1.978948e+04	1.051989e+00
min	1.000000e+00	1.000000e+00	5.000000e-01
25%	3.439500e+04	9.020000e+02	3.000000e+00
50%	6.914100e+04	2.167000e+03	3.500000e+00
75%	1.036370e+05	4.770000e+03	4.000000e+00
max	1.384930e+05	1.312620e+05	5.000000e+00

```
In [44]: ratings['rating'].mean()
```

```
Out[44]: np.float64(3.5255285642993797)
```

```
In [45]: ratings.mean()
```

```
Out[45]: userId      69045.872583
          movieId     9041.567330
          rating       3.525529
          dtype: float64
```

```
In [46]: ratings['rating'].min()
```

Out[46]: 0.5

```
In [47]: ratings['rating'].max()
```

Out[47]: 5.0

```
In [48]: ratings['rating'].std()
```

Out[48]: 1.051988919275684

```
In [49]: ratings['rating'].mode()
```

Out[49]: 0 4.0
Name: rating, dtype: float64

```
In [50]: ratings.corr()
```

Out[50]:

	userId	movieId	rating
userId	1.000000	-0.000850	0.001175
movieId	-0.000850	1.000000	0.002606
rating	0.001175	0.002606	1.000000

```
In [51]: filter1 = ratings['rating'] > 10  
print(filter1)  
filter1.any()
```

0 False
1 False
2 False
3 False
4 False
...
20000258 False
20000259 False
20000260 False
20000261 False
20000262 False
Name: rating, Length: 20000263, dtype: bool

Out[51]: np.False_

```
In [ ]: # Data Cleaning: Handling Missing Data
```

```
In [52]: movies.shape
```

Out[52]: (27278, 3)

```
In [53]: movies.isnull()
```

Out[53]:

	movieId	title	genres
0	False	False	False
1	False	False	False
2	False	False	False
3	False	False	False
4	False	False	False
...	...	...	...
27273	False	False	False
27274	False	False	False
27275	False	False	False
27276	False	False	False
27277	False	False	False

27278 rows × 3 columns

```
In [56]: movies.isnull().any()
```

Out[56]: movieId False
title False
genres False
dtype: bool

```
In [57]: movies.isnull().any().any()
```

```
Out[57]: np.False_
```

```
In [59]: tags.shape
```

```
Out[59]: (465564, 3)
```

```
In [60]: tags.isnull().any().any()
```

```
Out[60]: np.True_
```

```
In [61]: tags.isnull().any()
```

```
Out[61]: userId      False  
movieId      False  
tag           True  
dtype: bool
```

```
In [ ]: #There are null values. Lets remove those null values.
```

```
In [63]: tags=tags.dropna()
```

```
In [64]: tags.isnull().any().any()
```

```
Out[64]: np.False_
```

```
In [65]: tags.shape
```

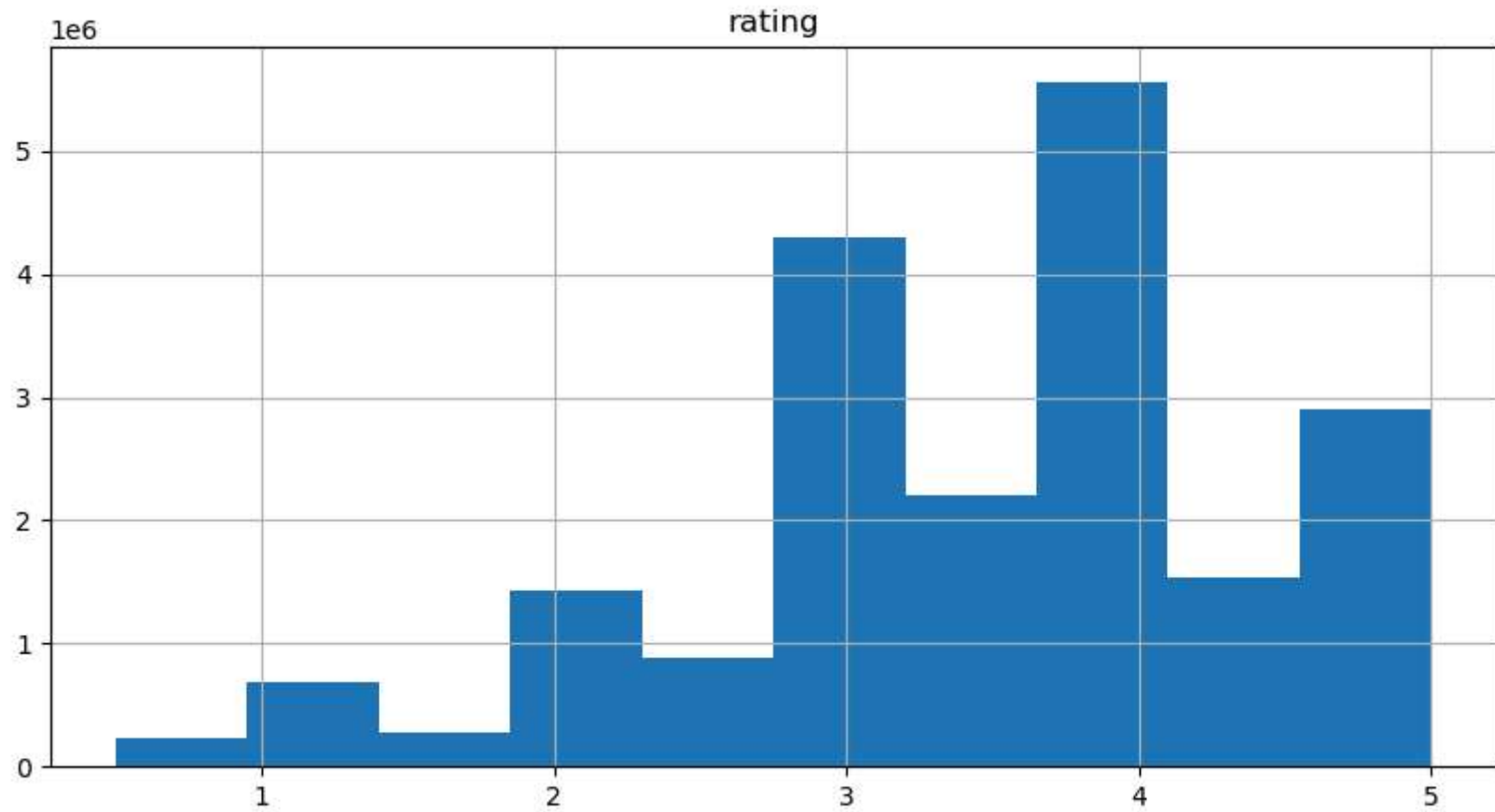
```
Out[65]: (465548, 3)
```

```
In [ ]: #Data Visualization
```

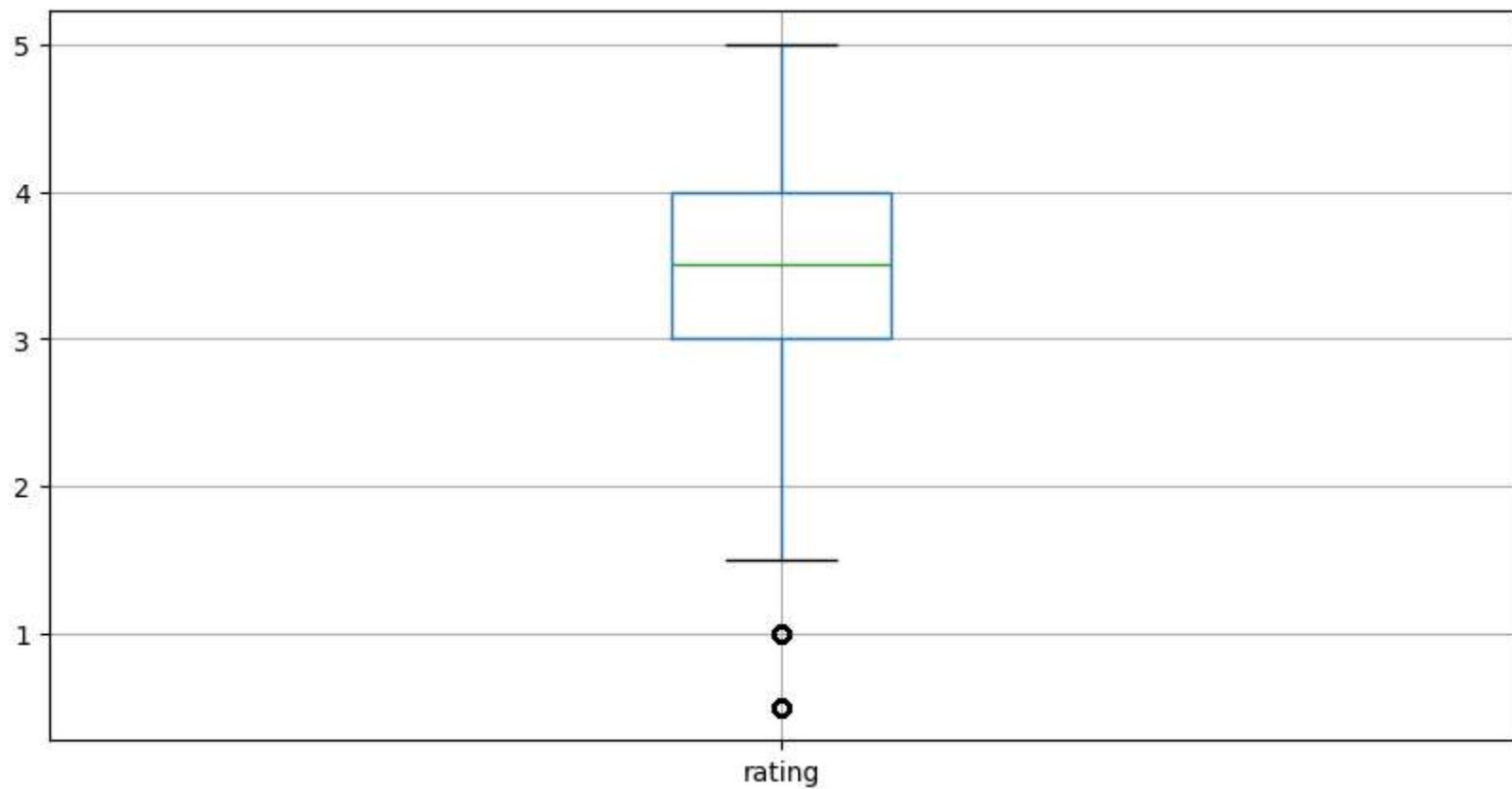
```
In [66]: %matplotlib inline  
ratings.hist(column='rating', figsize=(10,5))
```

```
Out[66]: array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```

```
In [67]: import matplotlib.pyplot as plt  
plt.show()
```



```
In [68]: ratings.boxplot(column='rating', figsize=(10,5))  
plt.show()
```



```
In [ ]: #Slicing
```

```
In [69]: tags['tag'].head()
```

```
Out[69]: 0    Mark Waters
1    dark hero
2    dark hero
3    noir thriller
4    dark hero
Name: tag, dtype: object
```

```
In [70]: movies[['title', 'genres']].head()
```

```
Out[70]:
```

	title	genres
0	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	Jumanji (1995)	Adventure Children Fantasy
2	Grumpier Old Men (1995)	Comedy Romance
3	Waiting to Exhale (1995)	Comedy Drama Romance
4	Father of the Bride Part II (1995)	Comedy

```
In [71]: ratings[-10:]
```

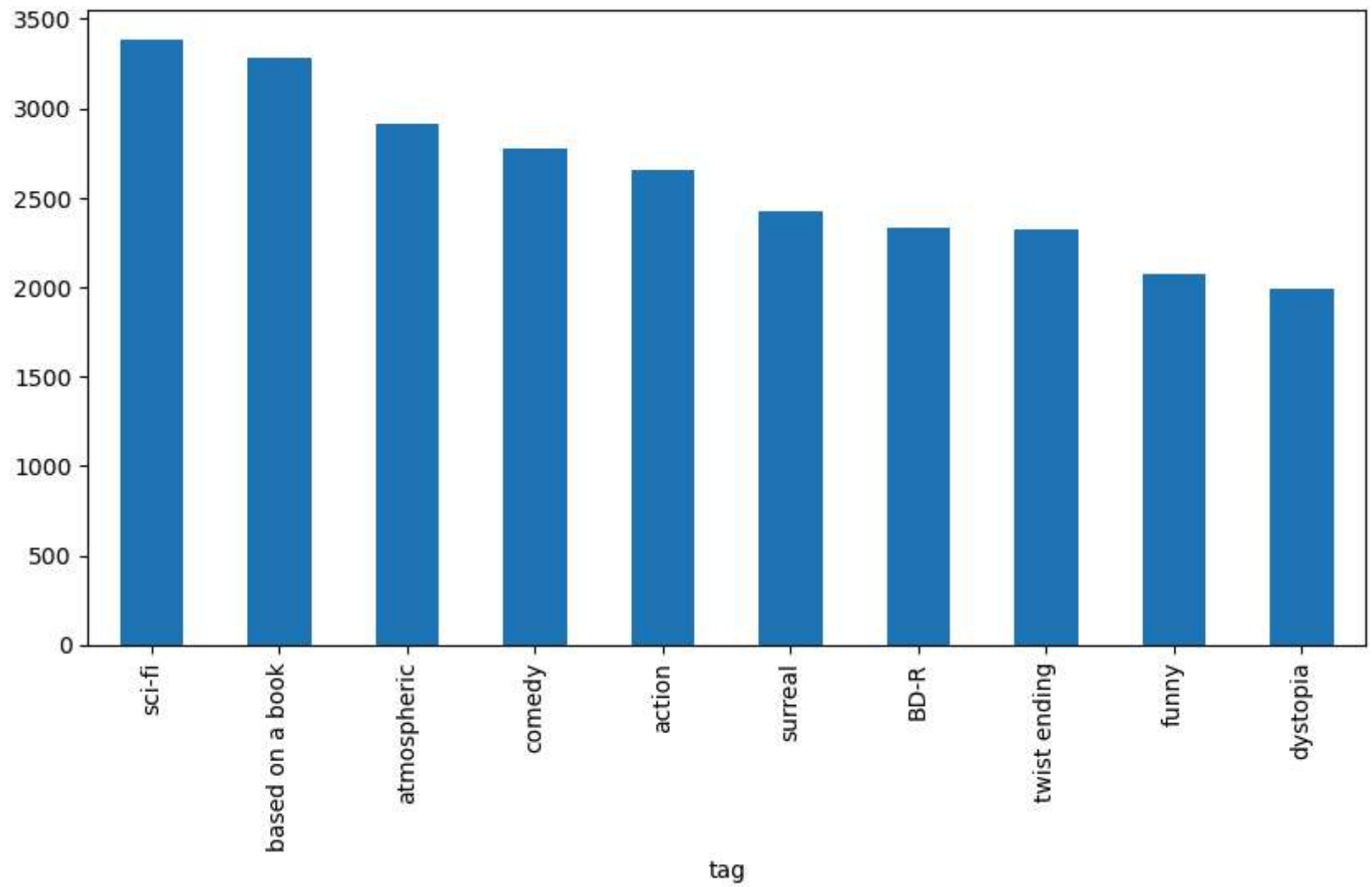
```
Out[71]:
```

	userId	movieId	rating
20000253	138493	60816	4.5
20000254	138493	61160	4.0
20000255	138493	65682	4.5
20000256	138493	66762	4.5
20000257	138493	68319	4.5
20000258	138493	68954	4.5
20000259	138493	69526	4.5
20000260	138493	69644	3.0
20000261	138493	70286	5.0
20000262	138493	71619	2.5

```
In [72]: tag_counts = tags['tag'].value_counts()
tag_counts[-10:]
```

```
Out[72]: tag
Hell naw 1
This is my happy face 1
I heel toe on Uday's house 1
Why? 1
Bobo 1
Diamond Dallas Page 1
I'm Devon Butler! 1
No argument 1
Really Bad 1
Botox 1
Name: count, dtype: int64
```

```
In [73]: tag_counts[:10].plot(kind='bar', figsize=(10,5))
plt.show()
```



In []: